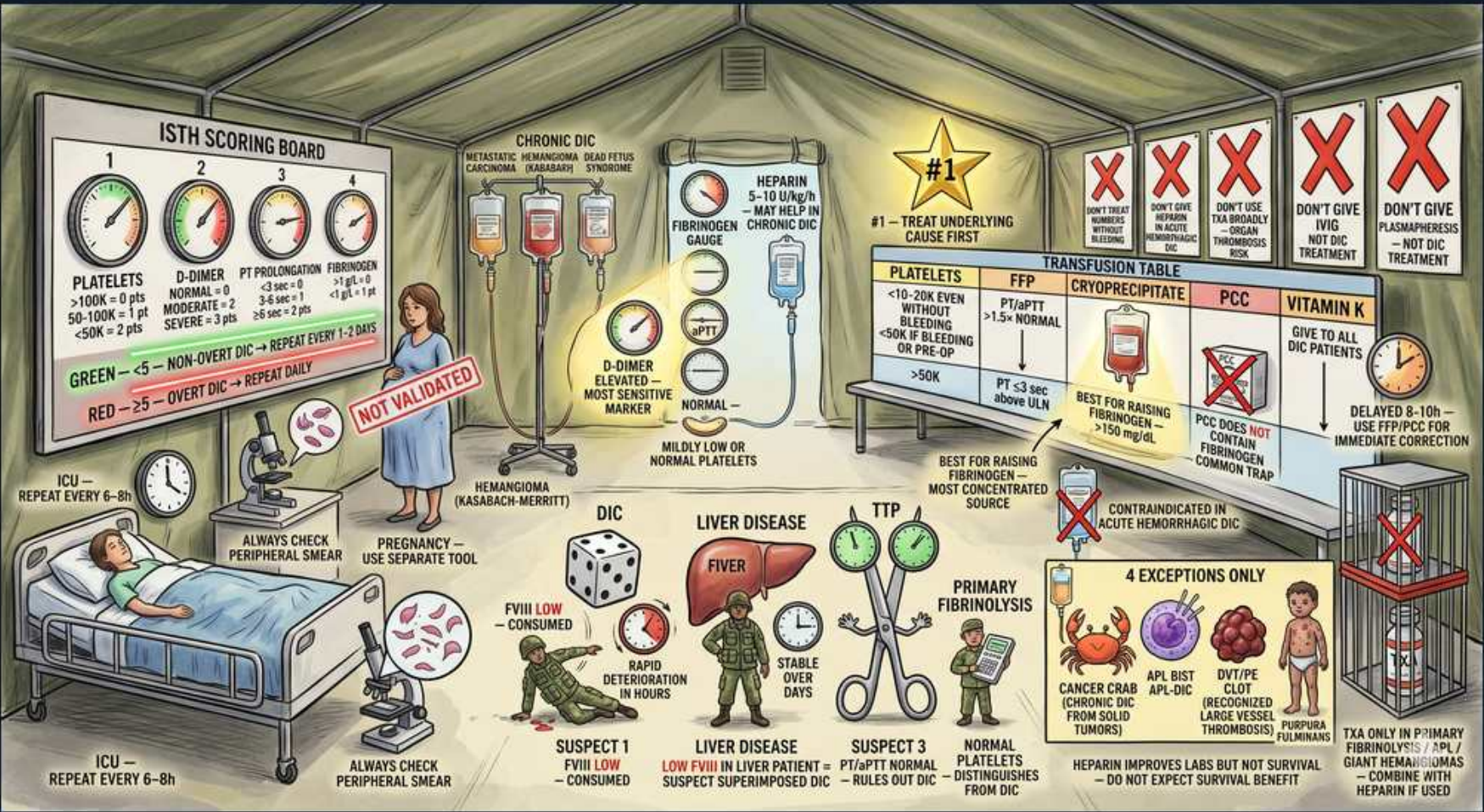


Coagulation — DIC (scoring, treatment, mimics)



1. ISTH scoring board

WHAT YOU SEE

Top-left 'ISTH SCORING BOARD' with four dial gauges (platelets, D-dimer, PT, fibrinogen) and a 'NOT VALIDATED' flag — DIC is a composite score, not one test

WHAT IT ENCODES

There is no single confirmatory test for DIC — diagnosis uses the ISTH composite score that points four parameters: platelet count, rise in fibrin marker (D-dimer/FDP), PT prolongation, and fibrinogen level. Each is scored 0–2 (D-dimer up to 3) and summed. The score is only validated when an underlying DIC-associated condition is present (note the image's 'NOT VALIDATED' flag — you must have a recognized trigger first).

BOARD TRAP

DIC is a CLINICAL + composite-lab diagnosis, never one value — a question that hangs the diagnosis on a single abnormal platelet count or D-dimer is wrong; and the ISTH score is invalid without an underlying predisposing condition.

2. Score >=5 = overt DIC

WHAT YOU SEE

The scoreboard color key 'GREEN <5 / RED >=5 overt DIC — repeat daily' — a cutoff that must be trended, not read once

WHAT IT ENCODES

An ISTH score ≥5 = overt (decompensated) DIC. A score <5 does NOT exclude DIC — it may represent non-overt/evolving DIC, so you repeat the score daily and watch the trend. Trending (falling platelets, falling fibrinogen, rising D-dimer over hours) is more informative than any single snapshot.

BOARD TRAP

A score under 5 never rules out DIC — re-score every 6–24h; the dynamic trajectory (a dropping platelet count day over day) is the real clue, especially in early sepsis or a retained dead fetus.

3. D-dimer most sensitive

WHAT YOU SEE

A center D-dimer gauge tagged 'most sensitive marker' — the lab that rules DIC out but not in

WHAT IT ENCODES

Elevated D-dimer/FDP is the most SENSITIVE laboratory marker of DIC, reflecting plasmin-mediated breakdown of cross-linked fibrin. A normal D-dimer makes active DIC very unlikely (good negative predictive value). It is the parameter weighted most heavily in the ISTH score (up to 3 points).

BOARD TRAP

D-dimer is sensitive but poorly SPECIFIC — it is elevated in VTE, recent surgery, trauma, pregnancy, liver disease and any inflammatory state, so a high D-dimer alone never proves DIC; use it to rule out, not to rule in.

4. Fibrinogen low (or normal early)

WHAT YOU SEE

A center fibrinogen gauge reading 'low / normal early' — an acute-phase reactant that can look falsely normal while being consumed

WHAT IT ENCODES

Fibrinogen is consumed in DIC, but because it is an acute-phase reactant it can start very high in sepsis/inflammation and therefore read 'normal' early even as it is being consumed. A falling fibrinogen trend is the meaningful finding; a fibrinogen <100 mg/dL is a strong sign of significant consumption.

BOARD TRAP

A 'normal' fibrinogen early does NOT rule out DIC — because it rises as an acute-phase reactant, a normal value may actually represent a large drop from a baseline of 600–800; always interpret the trend, not the absolute number.

5. Treat underlying cause first

WHAT YOU SEE

Top-center gold star '#1 — treat underlying cause first' — the only definitive DIC therapy

WHAT IT ENCODES

The #1 priority and only definitive therapy is treating the precipitating cause — antibiotics + source control for sepsis, delivery/evacuation for obstetric DIC, ATRA/chemo for malignancy, debridement for crush/burn. Everything else (products, heparin) is supportive and temporizing.

BOARD TRAP

Blood-product replacement alone will never fix DIC if the trigger persists — the fire keeps burning; the highest-yield answer to 'next best step' in DIC is almost always 'treat the underlying cause.'

6. Don't treat numbers if no bleed

WHAT YOU SEE

Top-right red Xs 'don't give if not bleeding — not a treatment' — products are for bleeding, not for normalizing labs

WHAT IT ENCODES

Transfusion in DIC is indicated only for ACTIVE bleeding or before an invasive procedure/surgery — not to normalize abnormal labs in a stable, non-bleeding patient. Transfusing into ongoing consumption can theoretically 'feed the fire.' Note also vitamin K is given to all DIC patients (it's cheap and corrects any coexisting deficiency) but is not a treatment for the DIC itself.

BOARD TRAP

Chasing the labs — transfusing platelets/FFP just to fix a number in a non-bleeding patient — is a classic wrong answer; products are reserved for bleeding or pre-procedure.

7. Platelets for bleeding

WHAT YOU SEE

The transfusion table platelet row '>50k if bleeding / 10-20k prophylaxis' — thresholds aimed at hemostasis, not a normal count

WHAT IT ENCODES

Platelet transfusion thresholds in DIC: keep >50×10⁹/L in an actively bleeding patient or before surgery, and >20×10⁹/L (some say >10) for prophylaxis in the non-bleeding patient at high risk. The aim is hemostasis, not a 'normal' count.

BOARD TRAP

Do not transfuse platelets for an asymptomatic mildly low count — and remember thresholds rise (≥50k) only when there is bleeding or a planned procedure; reflexively topping platelets to normal is incorrect.

8. FFP for factors

WHAT YOU SEE

The FFP column 'PT/PTT >1.5x normal -> replace factors' — global multi-factor replacement for the bleeding patient

WHAT IT ENCODES

Fresh frozen plasma replaces ALL consumed coagulation factors and is indicated for the bleeding DIC patient with PT/aPTT >1.5× normal. It contains some fibrinogen too but in dilute form, so it's not the ideal fibrinogen source. Use it when multiple factors need replacing.

BOARD TRAP

FFP is for global, multi-factor replacement in active bleeding with prolonged PT/aPTT — it is NOT the right choice when the dominant problem is isolated hypofibrinogenemia (that's cryoprecipitate), and not for an asymptomatic prolonged PT.

9. Cryoprecipitate for fibrinogen

WHAT YOU SEE

The cryoprecipitate column 'best for raising fibrinogen, most concentrated source' — the volume-sparing fibrinogen booster

WHAT IT ENCODES

Cryoprecipitate is the most concentrated source of fibrinogen (also FVIII, vWF, FXIII) and is the preferred product to raise fibrinogen when it falls below ~100–150 mg/dL in a bleeding patient. It delivers fibrinogen in a small volume, avoiding the fluid overload of large-volume FFP.

BOARD TRAP

Use CRYOPRECIPITATE, not FFP, as the primary fibrinogen booster — the trap pairs a bleeding DIC patient with fibrinogen <100 and offers FFP as a distractor; cryo is the concentrated, volume-sparing answer.

10. Heparin selective use

WHAT YOU SEE

The chronic-DIC panel showing a heparin drip 5-10 U/kg/hr tagged 'thrombotic/chronic DIC only' — anticoagulation that interrupts ongoing consumption

WHAT IT ENCODES

Heparin (usually low-dose) is reserved for the thrombosis-predominant or chronic/compensated forms of DIC — e.g., malignancy-associated chronic DIC, retained dead fetus, large aortic aneurysm, and purpura fulminans — where stopping ongoing clotting outweighs bleeding risk. It is anticoagulation that paradoxically helps by interrupting consumption.

BOARD TRAP

Heparin is NOT routine and is contraindicated in bleeding-predominant acute DIC (it worsens hemorrhage) — only the thrombotic/chronic subtypes benefit; giving heparin to an oozing acute-DIC patient is a wrong answer.

11. DIC vs liver: FVIII low

WHAT YOU SEE

Bottom DIC panel: a soldier collapsing labelled 'FVIII LOW — consumed' — in DIC factor VIII is eaten up along with everything else

WHAT IT ENCODES

Factor VIII is the discriminator between DIC and liver disease. In DIC, FVIII is LOW because it is consumed along with everything else. In liver disease FVIII is NORMAL or HIGH because it is synthesized by vascular endothelium (not just hepatocytes) and behaves as an acute-phase reactant. Both can show low platelets and prolonged PT/aPTT, so FVIII breaks the tie.

BOARD TRAP

A LOW factor VIII favors DIC over liver disease — the highest-yield distinguishing lab; if a question gives overlapping coagulopathy and you must separate DIC from cirrhosis, order/interpret the FVIII level.

12. Liver disease mimic

WHAT YOU SEE

Bottom liver panel 'low FVIII rules out DIC — stable over days' — liver coagulopathy keeps FVIII normal/high and evolves slowly

WHAT IT ENCODES

Liver disease mimics DIC with thrombocytopenia (splenic sequestration), prolonged PT/aPTT and low fibrinogen, but it is comparatively STABLE over days and has a NORMAL/HIGH FVIII. DIC, by contrast, evolves rapidly over hours with progressive consumption. Stigmata of chronic liver disease and a stable trajectory point away from DIC.

BOARD TRAP

Normal/high FVIII with coagulopathy points to LIVER disease, not DIC — and the time course helps: rapid evolution = DIC, slow/stable = liver; don't call a cirrhotic's stable coagulopathy 'DIC.'

13. TTP & primary fibrinolysis

WHAT YOU SEE

Bottom scissors/clock panels: TTP 'normal platelets...' and primary fibrinolysis 'normal D-dimer rules out DIC' — mimics separated by the rest of the panel

WHAT IT ENCODES

TTP/HUS produce schistocytes and thrombocytopenia like DIC, but PT, aPTT and fibrinogen are NORMAL (it's a platelet-VWF microangiopathy, not factor consumption). Primary hyperfibrinolysis (e.g., from prostate surgery, liver disease, t-PA excess) gives elevated FDPs but characteristically a NORMAL platelet count and normal/near-normal D-dimer because there is no cross-linked-fibrin clot being broken down.

BOARD TRAP

Schistocytes alone ≠ DIC — in TTP/HUS the coags and fibrinogen are normal; and in PRIMARY fibrinolysis the platelets and D-dimer stay normal while FDPs rise (normal D-dimer effectively rules out DIC). Use the whole panel, not the smear alone.

14. 4 exceptions: heparin first

WHAT YOU SEE

Bottom-right '4 EXCEPTIONS ONLY' box: cancer crab, APL blast, dead-fetus/DVT-PE clot, and purpura-fulminans infant — the settings where anticoagulation/specific therapy is used

WHAT IT ENCODES

There are four classic 'exception' settings where anticoagulation/specific therapy rather than pure bleeding-support is used: (1) malignancy-associated chronic DIC (heparin/LMWH), (2) APL (ATRA ± antifibrinolytics, cautious), (3) retained dead fetus syndrome (heparin to stabilize then deliver), and (4) purpura fulminans (heparin + protein C). In APL, tranexamic acid may be added cautiously for the hyperfibrinolytic bleeding.

BOARD TRAP

These heparin/specific-therapy exceptions apply ONLY to the chronic/thrombotic scenarios listed — do not extrapolate to ordinary acute bleeding-predominant DIC, where heparin and antifibrinolytics are harmful.